Technical Review

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1 Introduction

In this document, I will describe the code I have written for my ”Longitudinal analysis of heart rate variability as it pertains to anxiety and readiness” along with describe challenges I faced and what methods I elected to use. I will also reflect on the original scope I defined in the project proposal and how that scope changed throughout my milestones and final draft.

2 Code Description

I used the Data Spell IDE, a relatively new IDE (to my knowledge), catering to data analysts. I first created a scratch file, scratch.py, and began drafting the visuals (mainly barplots) that I would need, after data loading, cleaning and merging of course. However, this code soon became cluttered, so I instead made a new Notebook file, csys387-final-project.ipynb (still in Data Spell). This helped to organize my code into sections with text before each section briefly describing the following code chunk. I then made a second Notebook file, csys387-final-project-functions.ipynb, which included some of the function for statistical tests and sentiment analysis.

Some relevant packages I used were the NumPy and Pandas libraries. I primarily used the Pandas library to manipulate the data. I also used a lot of the Matplotlib library (ocasionally in tandem with Seaborn and tabulate) to create the visualizations in the report. Scipy also proved to be important with 3 or 4 different statistical tests and correlations used. I used re and TextBlob primarily for sentiment analysis.

2.1 Best Practices

- Readable... my code is well commented and includes descriptions in organized markdown cells.
- Modular... my code, specifically functions and libraries, are stored in their own file to clean up the look of my main file and to allow for use in other files upon import.

- Version control... I didn't use formal version control like GitHub, but I did have a spare Python file, old.py, where I stored discarded code. It was not organized and was not the most efficient way, but I did salvage the exercise type plot (in my final report Supplemental section) from old.py after discarding it without thinking.

3 Challenges

During the project process, I encountered several engaging challenges. One of the main difficulties was figuring out the story or journey that I wanted to take the reader on. Initially, I was torn between comparing the tags to one another, finding connections and building from there, or even having a few separate analyses that didn't build off of one another at all. It took some time to decide on a path, and I eventually elected to connect the autonomic nervous system to anxiety and compare the biometric variables that determine ANS balance to anxiety, emphasizing that connection. From there, I wanted to explore the specific tags and how they connected to the biometrics that determine anxiety and ANS balance. Although the final report didn't follow this plan exactly, the framework is there.

I also encountered smaller coding challenges, such as small errors during data frame merging or getting stuck trying to add text right above a bar plot. One detail that caused me trouble for a couple of hours was trying to plot asterisks over a correlation matrix to denote the statistically significant relationships.

Selecting the correct statistical test or coefficient for each combination was also a point of trouble for me. This required additional research, but it ultimately made me a better statistician and gave me a better understanding of my data. For example, I had never hear of Point Biserial before and had previously thought it was acceptable to use standard Pearsons to get correlation between binary variables.

Finally, I faced some ethical challenges when using my own personal data. My data contains information on the beat of my heart, my breath, and 60 variables describing how I live, along with my own personal journal where I enter what is going on in my life in brevity. I had to consider what to include and what to redact due to personal embarrassment. In this iteration, I chose to redact some variable names, but I hope to publish this work and remove the redacted sections for the sake of science!

4 Methods

- Point Biserial corr. coef.

  for binary to continuous correlations.

- Pearson's corr. coef

  for continuous to continuous correlations.

- Chi-squared contingency

  for binary to binary correlations.

- Two-sample t-test

  for testing difference of means hypotheses.

- TextBlob

  for sentiment analysis.

5 Scope

The original scope of my project included everything I have in my report currently, with the addition of a stepwise regression model to predict Morning Readiness. I had intended to try to recreate the model that EliteHRV uses to generate the Morning Readiness score. I hoped this would shed light on what matters for health (or at least what EliteHRV thinks matters).

I spent some time building a model with sklearn, mlxtend, and stats models (all still in csys387-final-project-functions.ipynb); however the model results in very poor performance when I tested it. I tried with various parameters, but the predictions were always poor, leaving an R2 score rarely above 0.1. I then also realized that the model building portion of my project may stick out in a bad way. It is very different from the traditional analysis of the other RQ's and so I omitted it. I feel this was the correct decision, as I think that model building process could or should have been its own research.